

Assignment – Lists & Tuples

Instructions:

All assignment programs should first be written by hand in a notebook, and then implemented and executed in Jupyter Notebook.

Please include proper comments in the code to make it easy to understand and readable.

Question 1:

Complete the following steps:

- Create a list called `colors` with the values `"red"`, `"green"`, `"blue"`
- Print the first item in the list
- Change the second item to `"yellow"`
- Add `"purple"` to the end of the list using `append()`
- Remove `"red"` from the list using `remove()`
- Print the list

Solution:

```
# Create a list

# Print the first item

# Change the second item to "yellow"

# Add "purple" to the end

# Remove "red"

# Print the list
```

Question 2:

Complete the following steps:

- Create a list called `fruits` with values `"apple"`, `"banana"`, `"cherry"`
- Print the first item in the list
- Change `"banana"` to `"mango"`
- Add `"orange"` at the end

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- Remove "apple"
- Print the final list

Question 3:

Complete the following steps:

- Create a list called **numbers** with values **1, 2, 3**
- Print the list
- Add **4** using **append()**
- Add **5** and **6** using **extend()**
- Insert **0** at position **0**
- Print the final list

Question 4:

Complete the following steps:

- Create a list **numbers** with **1, 2, 3, 4**
- Print the list
- Loop through list and print each item
- Print each item **multiplied by 2**
- Count total elements using **loop**
- Print the count

Question 5:

Complete the following steps:

- Create a list **numbers** from **1 to 5**
- Print the list
- Create a new list of **squares** using **list comprehension**
- Create a list of **even numbers** using **list comprehension**
- Print both lists

Question 6:

Complete the following steps:

- Create a list **numbers** with **5, 2, 9, 1**
- Print the list
- Sort in **ascending order**

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- Print the list
- Sort in **descending order**
- Print the final list

Question 7:

Complete the following steps:

- Create a list **numbers** with 1, 2, 3
- Print the list
- Copy the list using **copy()**
- Add **4** to copied list
- Print both lists
- Verify original is unchanged

Question 8:

Complete the following steps:

- Create **list1** with **1, 2** and **list2** with **3, 4**
- Print both lists
- Join using **+** operator
- Print result
- Extend **list1** with **list2**
- Print final list

Question 9:

Complete the following steps:

- Create a list named students with the following data:
Hint: Each student record should be stored as a list inside another list.
- Print the Complete Nested List
- Access the First Student Record
- Print the second student's name.
- Print Only the Marks of the Second Student
- Write a loop through a nested list to print the student name and marks

Name Marks	
Rahul	85
Anita	92
Vikas	78

Question 10:

Complete the following steps:

- Create a tuple called **fruits** with the values **"apple", "banana", "cherry"**

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- Print the second item in the tuple
- Print the number of items using `len()`
- Unpack the tuple into three variables `a`, `b`, `c`
- Print the variable `b`

Question 11:

Complete the following steps:

- Create a tuple called `colors` with the values `"red"`, `"green"`, `"blue"`.
- Print the first item in the tuple.
- Try changing the second item to `"yellow"` and observe the result.
- Add `"purple"` to the tuple by creating a new tuple.
- Remove `"red"` from the tuple by converting it into a list and then back into a tuple.
- Print the final tuple.

Question 12:

Complete the following steps:

- Create a tuple containing 5 fruits.
- Print the last element of the tuple.
- Find the length of the tuple using `len()`.
- Count how many times a value appears in a tuple.
- Convert a tuple into a list.

Question 13:

Complete the following steps:

- Create a tuple named `marks` containing the following data:
- Print the Complete Tuple
- Print the second tuple from the nested tuple.
- Print the Subject Name `"Science"`
- Print the Marks of `"English"`

Subject Marks	
Math	85
Science	90
English	88